

I'm Carlos Pereira Santos

Professor Data Science and Artificial Intelligence

Prof. Carlos Pereira Santos is Professor of Applied Data Science and Artificial Intelligence at Breda University of Applied Sciences (BUas), the Netherlands. His research focuses on data-driven real-time systems, applied AI, game engine technology, and distributed computing. Alongside his technical work, he shapes the university's AI policy, infrastructure, and laboratory strategy, and leads applied AI research initiatives.



Personal Statement



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After 10 years in industry as Technical manager, I spend the last 15 as a researcher, educator, and team builder working at the intersection of Artificial Intelligence, Game Technology, and interactive digital experiences. My career has been guided by a consistent goal: to improve European competitiveness in Digital Entertainment and Creative Industries by establishing practice-oriented connections between academia and industry — finding research-based solutions to real problems, and educating professionals who make a meaningful difference.

After a decade leading the Cradle research team, I developed a research identity that bridges industry practice and academic rigour. My work spans embodied AI, serious games, virtual production, and extended realities, always grounded in real-world deployment and societal relevance. None of this has been achieved alone; every milestone reflects the collective effort of the researchers, students, and industry partners I have had the privilege of working with.

I am seeking a part-time role at TU/e as a deliberate bridge between an applied and a research university. My goal is to strengthen the connection between both institutions, anchoring joint research while bringing applied implementation expertise, lab infrastructure, and industry networks. I aim to allocate my time as follows: Research (30%), Education (20%), Impact (20%), Team & Leadership (30%).

My longer-term ambition is to contribute to a stronger, more innovation-driven European research ecosystem — one where applied sciences universities and research universities work in genuine complementarity, producing research that is both methodologically rigorous and relevant to industry and society. I am committed to contributing to Europe's digital sovereignty, ensuring that creative, AI, and design industries develop on European values of openness, responsibility, and human-centred innovation. This role also represents a deliberate shift in my career focus: from building and leading large applied research teams, toward deepening research quality and co-developing the institutional partnerships that will define the next period of my academic contribution.

Ultimately, what motivates me most is the opportunity to inspire and equip the next generation — because I firmly believe that the young researchers and students we work with today will be the ones solving tomorrow's most important challenges.

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Educator

Identity as an Educator

BUas operates within a Project-Based Learning model where students engage with real-world challenges across integrative blocks. My role as an educator goes beyond teaching, it involves designing the conditions in which meaningful learning can happen.

Education achievements

I have been involved in the design and development of several programmes within the Academy for AI, Games and Media: the Bachelor in Applied Data Science and Artificial Intelligence, the Master in Game Technology, and the forthcoming Master in Physical AI.

This work includes curriculum architecture, Intended Learning Outcomes definition, directly linking European projects (VPSN, AVENUE, VPSN2) to educational programme, and the development of lab facilities, including the AI Makerspace, XR Stage Lab, Motion Capture Studio, and Photogrammetry Studio that connects research with practice. Alongside this, I have built sustained relationships with industry partners, ensuring that the challenges students work on are shaped by real professional contexts.

Vision of Education

I believe Europe needs stronger innovation pathways where practitioners with deep industry knowledge can apply research methods to solve real problems. Universities of Applied Sciences are uniquely positioned to build these pathways, but only if education and research are genuinely mutually reinforcing: student projects feed into research questions, research outputs feed back into educational design, and industry partners co-shape both.

The BUas-TU/e partnership is central to realising this vision at scale. By combining BUas's applied, practice-oriented culture with TU/e's research rigour and doctoral infrastructure, we can create learning environments where students benefit from both institutions' expertise — and where supervision becomes a collaborative research act rather than an administrative responsibility.



Factsheet - Education

Educational model: BUas Academy for AI, Games and Media operates fully within a Project-Based Learning (PBL) model. The academic year is divided into 4 blocks, each with a single integrative student project worth 15 ECTS run by many lectures. Teaching activities take the form of project definition, advising on Intended Learning Outcomes (ILOs), and research mentorship rather than lecture-based delivery.

Course		Years	ETCS	Role
Master Media Innovation - Technical Module	MSc	2018-2022	15 ECTS	Responsible
Master Game Technology	MSc	2022-2024	60 ECTS	Lecturer

Graduation Year	PhD Supervision	MSc Thesis	BA Thesis
2025	2 (on-going)	0	1
2024	1 (on-going)	1	1
2023		2	1
2022		1	1
2021		3	0

Research

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Vibe - Digital Human



Maritime Spatial Planning Challenge



Procedural Ocean - Shipping routes



Avenue Report presentation

Identity as a Researcher & Vision of the Research Field

I am an applied researcher working at the intersection of Game Technology, Artificial Intelligence, and human-computer interaction. My research asks how intelligent, interactive systems can be designed to produce insight and impact in real-world contexts, from understanding human behaviour and experiences, to AI-driven tools for complex societal decision-making.

I research I do has multiple audiences in mind: publications for academic communities, frameworks and tools transferable to educational practice, and solutions directly usable by industry partners. This triple orientation reflects my conviction that applied sciences research is the critical bridge between fundamental inquiry and societal relevance.

Main Research Achievements (last 8 years)

Four key outputs demonstrate the breadth and applied impact of my research:

- Procedural Ocean (2023-2024): Led Horizon Europe project generating AI-driven 3D offshore environments based on GIS data for energy planning and decision-making. Published BUILD-IT 2023, COG 2026.
- PlanV (2021-2023): Virtual Productions applied research (w/ Painting Practice) using VR pre-visualisation built on Unreal Engine, enabling film and television production professionals to iterate on set design, cameras, lighting, and animation in real time, reducing production costs while increasing efficiency. Conceived deployed on His Dark Materials (BBC/HBO) and Wednesday (Netflix).
- VIBE (2018-2023): Led project developing photogrammetry-based digital human avatars and embodied conversational agents for medical simulation training. Published ACM IVA 2023, ACM IUI 2023.
- MSP Challenge (2016-2019): Co-developed multi-awarded serious game platform for Marine Spatial Planning across five international seas. Published in Journal of Ocean Technology and Ecology & Society.

Research Ambitions & Strategy (next 5-10 years)

My agenda continues to center on Human collaboration with Intelligent Systems, driven from the Game Technologies and Digital Twins, and I will explore how Physical AI can be embodied, grounded in physical environments, and support human decision-making.

The main driver, developing research on ethical by design AI, and the use of creative practices at the intersection of AI, design, and societal impact.

Factsheet - Funded Research Projects

Project	Funding	Dates	Role
VPSN2 - Virtual Productions Studio Network	Erasmus+	2025-2028	Principal Investigator
AVENUE - European ecosystems for Audiovisual Education and Training Network for European Vocational Excellence	Erasmus+	2025-2029	Principal Investigator
KreativEU - Long-term institutional, structural and strategic cooperation on Europe's cultural heritage - Sustainability of AI	European Alliance	2024-2027	Work Package Leader
VPSN - Virtual Productions Studio Network	Erasmus+	2023-2026	Principal Investigator
AICI - AI in the Creative Industry — Generative AI opportunities and challenges for Creative Digital Industry	Erasmus+	2024-2025	Principal Investigator
Illiad - Procedural Ocean - Auto-generated 3D environments for offshore energy planning and decision-making	Horizon	2023-2024	Principal Investigator & Technical Coordinator
VIBE - Realistic animated face avatars based on 3D photogrammetry for medical education	OPZuid / Regional	2018-2023	Principal Investigator & Technical Coordinator
MSP Challenge - Multi-awarded serious game for Marine Spatial Planning across 5 international seas	Horizon Europe	2016-2019	Technical Coordinator

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Impact

Identity as a Value Creator & Vision of Impact

I see impact as the natural endpoint of applied research, the moment when a tool, framework, or platform moves beyond the lab and changes how people work, decide, or learn. My impact spans three domains: industry practice (tools deployed in professional film and healthcare), policy and governance (serious game platforms used by marine planners and government agencies), and education and society (AI literacy frameworks and Virtual Productions Community adopted across Europe).

Main Achievements in Impact (last 8 years)

- VP Gathering: Founded and organised annual Virtual Production industry-education conference in Breda, for the last 6 years, and now bringing 250 participants bridging education, research and creative industry practice at national and international level.
- Development of advanced BUas Lab Facilities: Established and led development of the AI Makerspace, XR Stage, XR Lab, Motion Capture Studio, and Photogrammetry Studio — infrastructures available for education, research, and industry partners across BUas and the Netherlands.
- Educational Programme Support: Co-developed the Applied Data Science & AI (ADS&AI) BSc programme at BUas, and contributed externally to the development of obstetric simulation training through the VIBE project, and embedding research outcomes directly into professional medical education.
- Keynote presentations and Industry Partnerships: Established and maintains communication and strategic partnerships with Epic Games, NVIDIA, and working now with Google DeepMind, connecting BUas research and education to leading global technology organisations.
- KreativEU — leading the work across an alliance of 11 European universities, developing shared frameworks for responsible and sustainable AI in creative industries and cultural heritage contexts.

Ambitions & Strategy for Impact

My ambition is to establish BUas-TU/e collaboration focusing on applied AI impact in the Brabant region and beyond. This means co-developing an industry network where research questions are co-shaped by professional partners, outcomes are rapidly transferred into practice, and open science principles guide dissemination. I will continue pursuing funding that rewards societal relevance alongside academic rigour.



XR Stage Lab - Edu Workshop



AI Makerspace Lab - George



VP Gathering 2026



VP Gathering 2026

Factsheet - Impact

Activity	Dates	Role
Peer-review Publications	2002–Present	With over 40 peer-reviewed publications, focusing in IEEE and ACM conferences in the areas of AI, technical solutions, and machine human interaction.
Invited Speaker in International Events	2020–Present	Invited Speaker and Keynotes to several international over 40+ events: Sustainable Tourism Mobility Forum (Nov 2025), SURF.nl — Virtual Spaces: Behind the scenes (Sep 2024), LevelUp Conference (Aug 2023), AI Forward Forum & Google Developer (Mar 2023).
VP Gathering — Conference Organiser	2020–Present	Founded and led annual Virtual Production industry-education conference in Breda; now attracting 250+ national and international professionals.
BUas AI Strategy Implementation	2024–Present	Part of the group leading the BUas institutional AI strategy, implementation and innovation leadership.
Direct Industry Collaboration in Projects and Events	2020–Present	Directly collaborations with industry partners, including big brands: Epic Games, ROE Visual, Amazon, HP, Canon, Google, Sony, Chaos across 20+ applied research projects and events.
Development of Several Lab Facilities	2018–Present	Developed several Educational and Research infrastructures: Photogrammetry Studio (2018), XR Stage (2020), Motion Capture (2023) and AI Makerspace (2026).
Visiting Researcher, Griffith University, Australia	2023	Invited to New York Counterpoint short film production; exploring joint research and developing a Shared PhD protocol.

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Team and Leader



XR Lab + Internships 2025



Academy AI, Games and Media Research 2024



VPSN 2 - Workshop 2026

Identity as a Leader & Vision of Academic Citizenship

My identity is shaped by two decades of experience across industry and academia: as a technical and project lead on critical safety software for the European Space Agency, and as the founder and leader of multiple research teams at BUAs. I understand leadership from both sides: what it means to be an effective team member who contributes to shared goals, and what it takes to build a team from scratch and guide it to independence.

I want to be clear that none of my achievements are individual. Every project, programme, and platform I have been associated with is the result of collective effort — and I am genuinely appreciative of every person I have had the pleasure of collaborating with. I believe good leadership is about people development, but the growth of the individuals behind them, good outputs are a byproduct.

My role has been to create the conditions for that collaboration to thrive: designing the structure, recruiting the right people, establishing shared culture and working agreements, and stepping back as teams like Cradle and the XR Stage became self-sustaining, and now I am doing the same with the AI Makerspace.

Leadership Development Ambitions

My leadership ambitions operate at three levels.

At **team** level, I aim to continue developing research leadership capacity within BUAs, mentoring junior researchers into PI roles and supporting staff pursuing doctoral training at TU/e.

At **institutional** level, I am working directly with the Academy Management Team and BUAs Executive Board on AI strategy and research governance, and aim to formalise the BUAs-TU/e partnership into a sustainable collaborative structure with shared supervision and joint industry engagement.

At **European** level, through consortia, I will continue building leadership roles in research projects and alliances that advance responsible AI and practice-oriented research across the continent.

Factsheet - Team

Activity	Dates	Role
Data Science and AI R&D Team	2025-Present	Manager: Started to developed a new R&D group now with 3 FTEs focus on Artificial Intelligence research. I as involved in the hiring procedure, deadlines, quality, planning and task division.
AI Makerspace Team	2026-Present	Manager: Successfully developed, and managed a group of 1 FTEs (4 persons) focus on coordination and management of the Physical AI Lab.
BUas AI Pioneer Team	2024-Present	Member of BUas institutional AI strategy and innovation leadership team.
XR Stafe Team	2020-2025	Manager: Successfully developed, and managed a group of 1.2 FTEs (3 persons) focus on coordination and management of the XR Stage and MoCap Studio.
Cradle R&D Team	2016-2025	Responsible: Successfully developed, and coordinated a very qualified R&D group (cradle.buas.nl) now with 20 FTEs focus on research linked with Creative Industries. I as involved in the hiring procedure, deadlines, quality, planning and task division.

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Thank you

I have been building bridges, between education, research and practice, between people, and I am looking forward to contributing the energy, curiosity, and collaborative spirit to drive positive impact and solving important challenges.

Email : *santos.c@buas.nl*